Modulation of Autophagy by High Glucose and Metformin in Mouse Retinal Explants

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BACKGROUND: Diabetic retinopathy (DR) is a major complication of diabetes mellitus and the leading preventable cause of blindness worldwide (Seewoodhary, 2021). This condition is driven by chronic hyperglycemia which induces oxidative stress and inflammation, leading to microvascular and neuroretinal alterations (Al-Kharashi, 2018). Despite being considered a microvascular disease, increasing evidence suggests that neurodegeneration and neuroinflammation are key contributors to DR pathogenesis (Che et al., 2026). Autophagy, the main intracellular pathway responsible for degrading organelles and macromolecules (Palmer et al., 2025), is a known player in retinal diseases (Adornetto et al., 2020), yet its role in DR remains unclear. Nevertheless, several studies have shown that pharmacological manipulation of autophagy exerts neuroprotective effects in the retina (Satriano et al., 2025). In this study, we investigated autophagy changes in mouse retinal explants exposed to normal and high-glucose conditions and evaluated the effects of the direct exposure of retinal explants to metformin, a glucose-lowering agent known for its autophagy-modulating and neuroprotective properties.

METHODS: Organotypic retinal cultures from C57BL/6J mice (8-10 weeks) were exposed to normal (15mM) (NG) or high (50mM) glucose (HG) for 7 days or maintained for 4 days in vitro (DIV) in the presence or absence of metformin (1µM or 10µM). Neuronal survival and glial activation were analyzed by immunofluorescence. Protein and mRNA levels of autophagy-related proteins (LC3, p62, p-ULK1, LAMP1, p-AMPK) and inflammatory markers (IL-1b, GS, GFAP) were analyzed by western blotting or qPCR.

RESULTS: Retina exposed to HG for 7 day exhibited loss of neuronal components and increased glial reactivity compared to the ones cultured under NG conditions. HG treatment was associated with the upregulation of inflammation-related proteins and significant changes in autophagic flux.

Metformin modulated autophagy-related markers in retinal explants maintained for 4DIV and counteracted the increased glial reactivity markers compared to baseline (DIV0).

CONCLUSIONS: Our preliminary data suggest that autophagy is significantly modulated in the retina under HG conditions and can be directly targeted by metformin. These finding provides a rational for future studies aimed at pharmacologically modulating HG-induced autophagy to clarify its role in the pathogenesis of diabetic retinopathy